C

Meet memory and the machine

THE MENTAL MODEL

You specify small operations and must watch every memory boundary.

SOURCE → COMPILE → MEMORY → PROGRAM

READ THE CODE

```
#include <stdio.h>
int main(void) {
  int x[3] = {10, 20, 30};
  int total = x[0] + x[1] + x[2];
  printf("%d\n", total);
  return 0;
}
```

FOUR CLUES TO READ IT

- #include adds declarations from a header.
- int is an integer type; x[0] is the first element.
- printf prints formatted text.
- main is the program entry point here.

WHAT MAKES IT DIFFERENT

- Explicit memory addresses and pointers enable precise low-level work.
- Compiled native programs are common in operating systems and embedded devices.

WHERE IT FITS

Firmware, operating-system components, hardware-adjacent code.

WATCH OUT

Out-of-bounds access can corrupt memory; the programmer must prevent it.

TRY ALONE • 60 SEC

Why is x[3] invalid for an array with three elements?